GasTos PH: A Web-Based Fuel Cost Calculator and Live Price Tracker for Philippine Drivers

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Abstract

This report presents the design and development of GasTos PH, a client-side web application that provides Filipino drivers with real-time fuel cost estimation, live DOE-sourced price data, and interactive station mapping. The system integrates four core modules — a multi-mode fuel cost calculator, a regional price dashboard, an OpenStreetMap-based station locator, and a community price reporting interface — delivered as a single-page application built entirely with HTML5, CSS3, and vanilla JavaScript. The calculator supports vehicle-type-aware fuel filtering, road and straight-line distance estimation via OSRM and Leaflet, and multi-brand station price comparison using DOE weekly advisory data. Results demonstrate accurate trip cost estimation across all Philippine regions with near-zero latency on modern browsers. The application addresses the practical need for accessible, up-to-date fuel cost information amid record-high pump prices in the Philippines as of April 2026.

Keywords: Fuel Cost Calculator, Web Application, DOE Philippines, OpenStreetMap, Leaflet.js, Vanilla JavaScript, Single-Page Application

1 Introduction

The Philippines has experienced significant fuel price surges in 2026, driven by global supply disruptions stemming from Middle East geopolitical tensions. As of the DOE weekly price advisory for April 22–28, 2026, RON 95 gasoline in Metro Manila averages ₱88.10/L — a record high that places considerable financial pressure on everyday drivers, delivery workers, and commuters who rely on personal vehicles.

Despite the availability of DOE weekly advisories, there is no widely accessible tool that aggregates this data, maps nearby stations, and directly computes trip fuel costs for Filipino users in a single interface. Existing general-purpose fuel calculators do not account for Philippine-specific fuel grades (RON 91, RON 95, RON 97/100, Diesel, Kerosene), local station brands (Petron, Shell, Caltex, Seaoil, Phoenix, Flying V, Cleanfuel, Jetti), or regional price variation across the country's 17 administrative regions.

GasTos PH was developed to fill this gap — a lightweight, installable-free web application that consolidates fuel price lookup, trip cost calculation, station discovery, and community price reporting into one dark-themed, mobile-responsive interface.

2 Objectives

The project was guided by the following primary objectives:

- Design and implement a multi-mode fuel cost calculator supporting Simple, Advanced, and Vehicle Comparison modes
- Integrate DOE weekly fuel price data across all 17 Philippine regions and 8 major station brands
- Build an interactive map module using OpenStreetMap/Nominatim for location search and OSRM for road distance estimation
- Implement a station locator using the Overpass API to surface nearby fuel stations within 3 km of any searched location
- Develop a vehicle-type-aware fuel filtering system that restricts fuel type options based on the selected vehicle
- Provide a community reporting module for crowd-sourced pump price submissions

3 Methodology

3.1 System Architecture

GasTos PH is a fully client-side single-page application (SPA) with no backend server or database. All application logic runs in the browser. The codebase is modular, split across six JavaScript files with clearly separated concerns:

File	Responsibility
data.js	Static fuel price data (DOE weekly) and brand price tables
ui.js	DOM rendering, navigation, ticker, toast notifications
map.js	Leaflet calculator map, station locator, routing
autocomplete.js	Nominatim-powered location search with dropdown
calculator.js	Fuel cost logic, vehicle-fuel mapping, station comparison
main.js	Application entry point and initialization

Table 1. Codebase for GasTos PH

3.2 Data Sources

Fuel price data is sourced from the DOE Philippines weekly price advisory and stored statically in data.js as structured JavaScript objects. Prices are available for all 17 regions plus NCR, covering RON 91, RON 95, RON 97/100, Diesel, Diesel Plus, and Kerosene. Brand-specific prices for 8 major stations are stored separately in a BRANDS array and used for station price comparison.

Map data, geocoding, and station discovery rely entirely on open APIs: OpenStreetMap/Nominatim for location search and reverse geocoding, OSRM for road routing and distance calculation, and the Overpass API (with three mirror fallbacks) for querying nearby fuel station POIs.

3.3 Calculator Module

The calculator offers three modes. Simple mode accepts origin/destination (via map or manual input), fuel type, gas station, fuel price, and fuel efficiency (km/L), then computes total trip cost, liters consumed, and optionally doubles the distance for round trips. Advanced mode adds vehicle type presets (motorcycle ~35 km/L, sedan ~12 km/L, SUV ~9 km/L, van/truck ~7 km/L, hybrid ~22 km/L, jeepney ~4 km/L) and passenger cost splitting. Compare mode allows two vehicles to be evaluated on the same route side by side.

3.4 Vehicle-Type Fuel Filtering

A VEHICLE_FUEL_MAP object maps each vehicle category to its compatible fuel types based on standard Philippine vehicle fuel usage:

Vehicle	**Compatible Fuels**
Motorcycle	RON 95, RON 91
Car	RON 95, RON 91, RON 97/100
Van / AUV	Diesel, RON 95, RON 91
Truck	Diesel
Jeepney / Bus	Diesel
Hybrid	RON 95, RON 91

Table 2. Vehicle Fuel Map for GasTos PH

When a vehicle is selected, the fuel type dropdown is dynamically rebuilt to show only compatible options, preventing invalid fuel-vehicle combinations.

3.5 Map and Station Locator

The calculator map uses Leaflet.js with CARTO Dark Matter tiles. Users can search a Philippine location via Nominatim autocomplete, click the map to pin origin/destination, or drag markers to fine-tune positions. Road distance is computed via OSRM; straight-line distance uses the Haversine formula with a 1.35 road-factor multiplier as an estimate.

The station locator queries the Overpass API for fuel amenity nodes within 3 km of a searched location. Three Overpass mirrors are tried sequentially with a 10-second timeout per mirror, with a Nominatim-based fallback if all mirrors fail. Results are deduplicated by coordinate and name, sorted by distance, and capped at 15 stations. Matched stations display brand-specific DOE prices in the detail panel.

3.6 UI and Interaction Design

The interface uses a dark theme with CSS custom properties for consistent theming. Navigation is tab-based across four pages: Calculator, Prices, Map, and Community. A live price ticker scrolls NCR fuel prices across the top of the application. Toast notifications provide non-blocking feedback for user actions. All interactive elements (dropdowns, toggles, map interactions) update the UI reactively without page reloads.

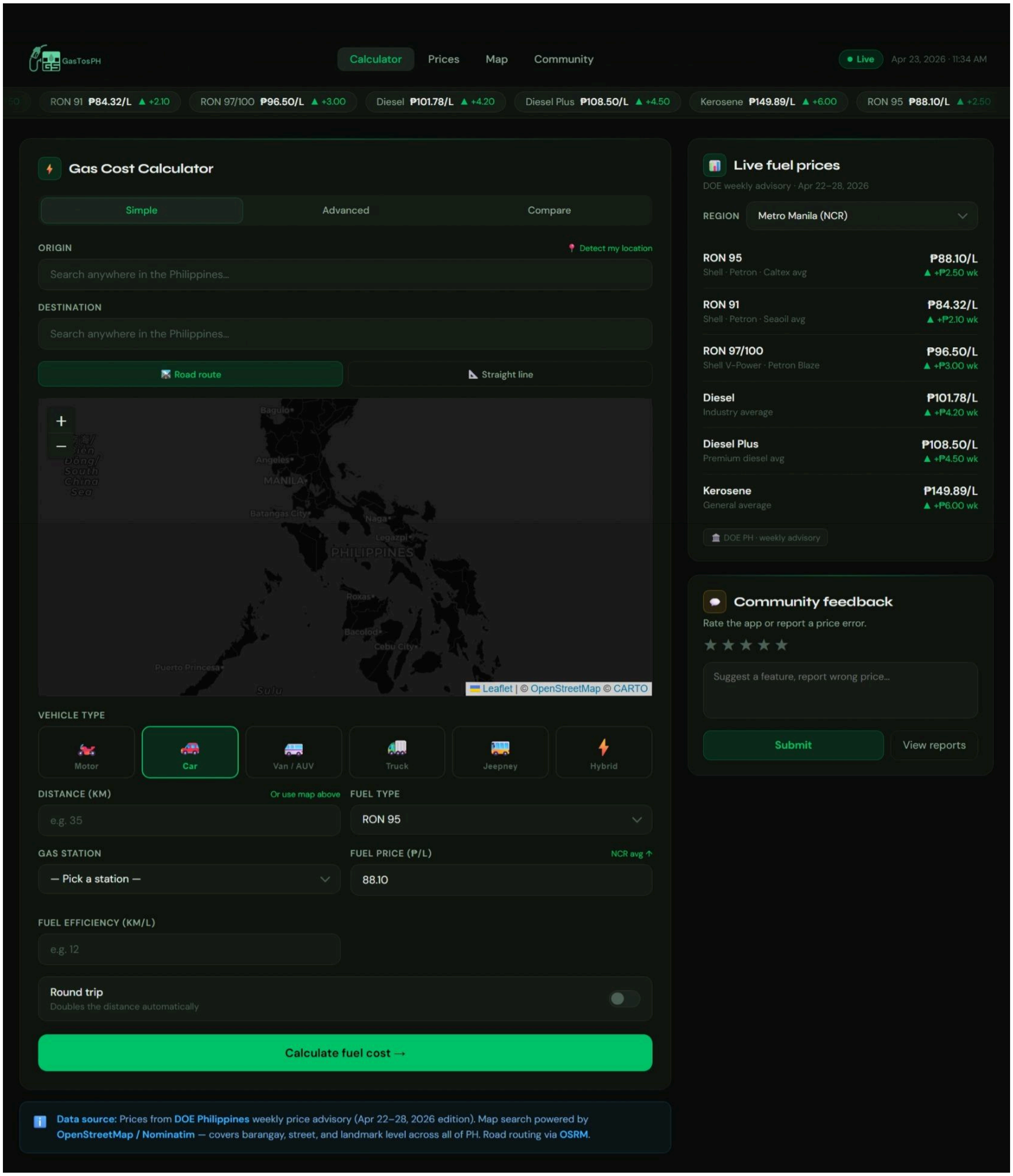

Figure 1. GasTos PH Gas Cost Calculator, with Live Fuel Prices and Community feedback page

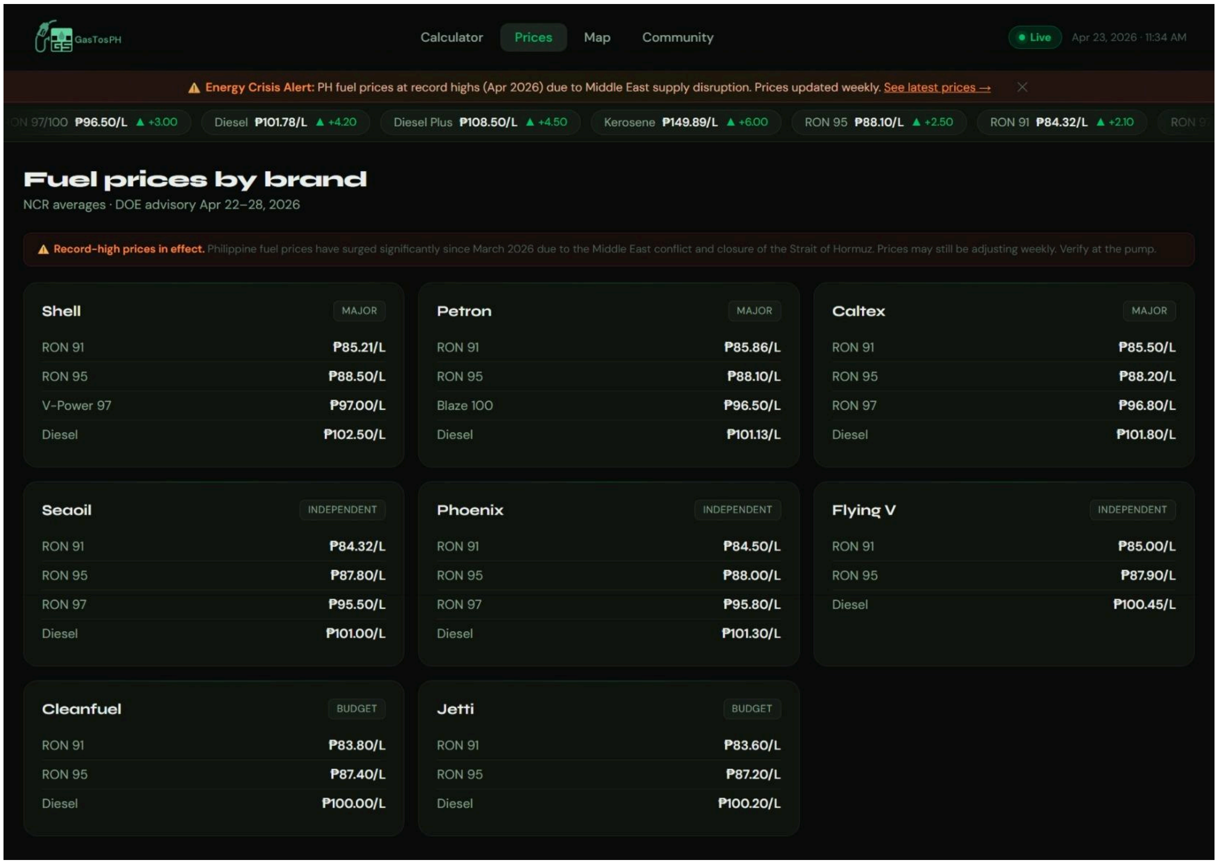

Figure 2. Prices page

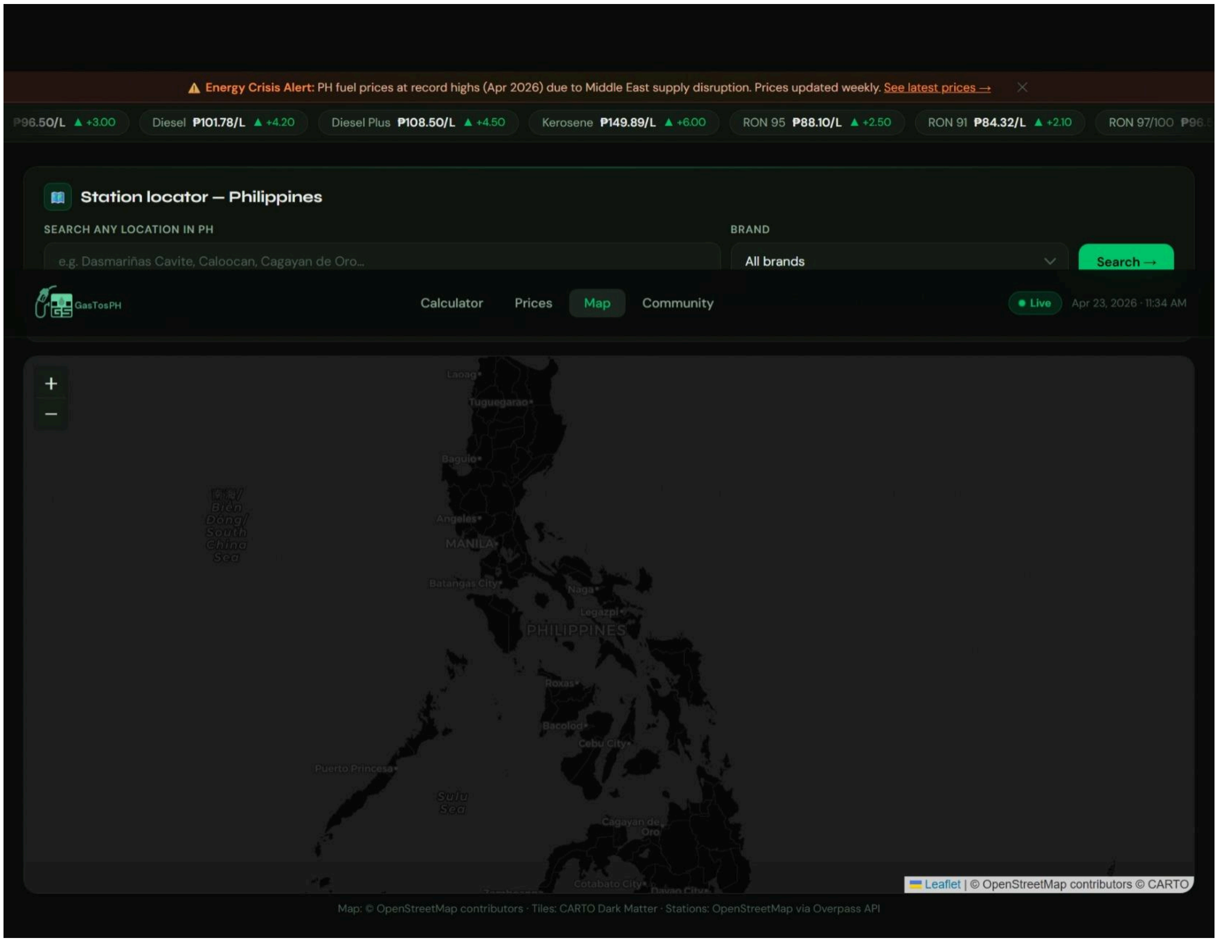

Figure 3. Map page

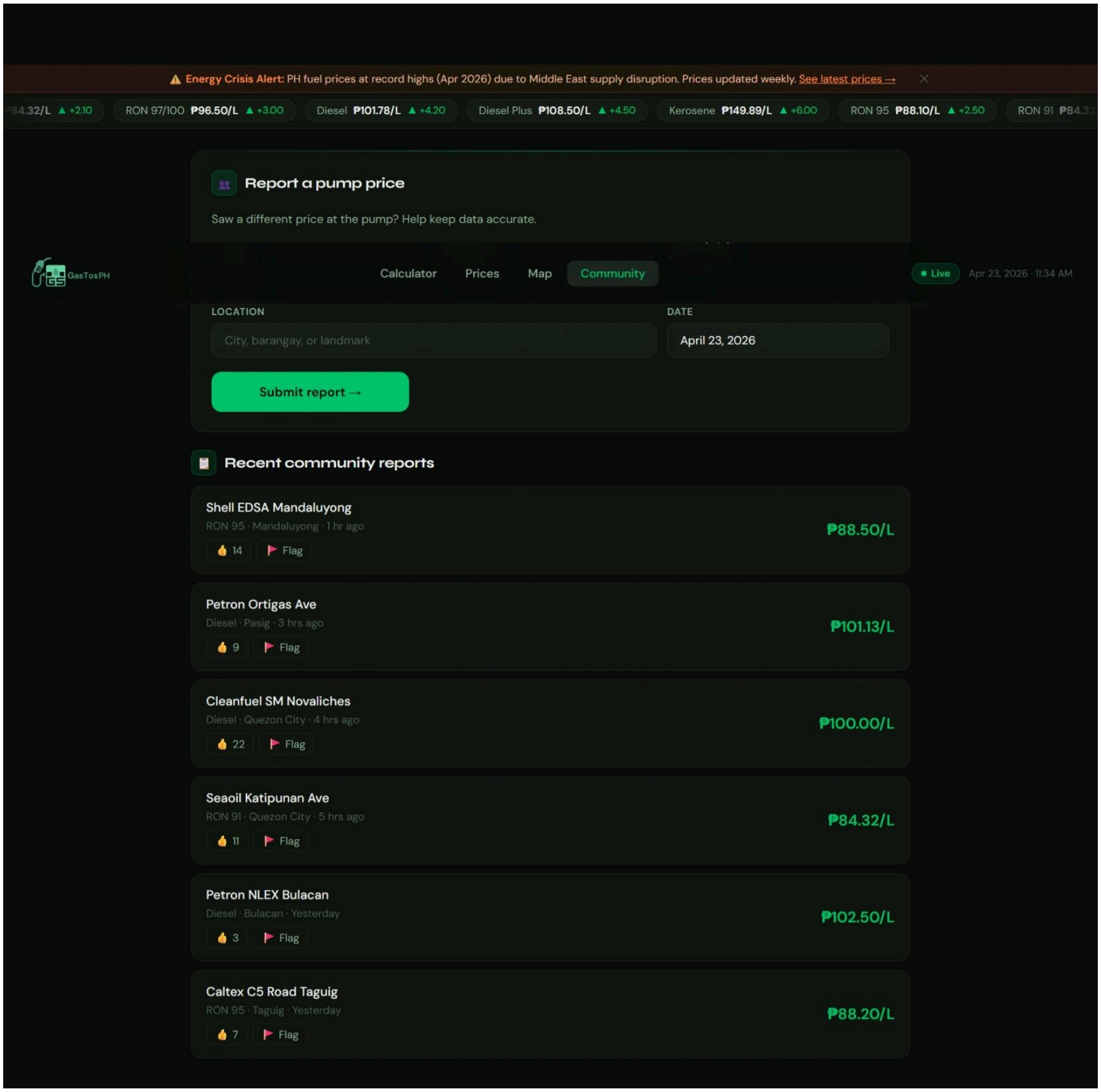

Figure 4. Community page

4 Results and Analysis

The application successfully implements all stated objectives. The fuel cost calculator produces accurate results across all three modes, with trip cost calculated as: Cost = (Distance / km/L) × Price per Liter. The vehicle picker correctly filters fuel options per vehicle type, and the station comparison panel ranks all 8 brands by price for the selected fuel type, highlighting savings versus the cheapest available option.

The map module resolves location searches across barangay, street, and landmark levels throughout the Philippines. Road routing via OSRM successfully returns distances for most urban and inter-city routes; routing failures gracefully fall back to straight-line estimates. The Overpass station locator returns results within 2–4 seconds under normal network conditions, with mirror fallback adding up to 10 additional seconds in degraded conditions.

Regional price data is accessible for all 17 Philippine administrative regions. The live price ticker and price dashboard reflect the current DOE weekly advisory, and the "Use ↑" function in the station comparison panel correctly pre-fills the calculator with the selected station's price (this button is intentionally absent from the Map panel and Live Prices sidebar to avoid redundant navigation flows).

5 Discussion

GasTos PH demonstrates that a fully functional, data-rich web utility can be delivered without a backend server, database, or build pipeline — relying entirely on static data, open APIs, and browser-native capabilities. This architecture makes the application trivially deployable (a single folder of files), fast-loading, and free to host on platforms such as GitHub Pages.

From a Human-Computer Interaction perspective, GasTos PH addresses a real usability gap in how Filipino drivers access fuel cost information — consolidating what was previously a multi-step process of reading DOE PDF advisories, cross-referencing fuel types, and manually computing costs into a single, task-oriented interface. Several deliberate HCI decisions support this: the vehicle type picker constrains fuel options to only what is valid for the selected vehicle (applying the HCI principle of error prevention), the map-integrated distance input removes the need for users to know exact kilometer values, and toast notifications provide immediate feedback for every action. As HCI literature emphasizes, usability is not just about aesthetics but about whether a system allows users to complete tasks quickly and with minimal frustration — and in the context of a fuel price crisis where every peso matters, reducing the cognitive load of cost estimation from minutes to seconds demonstrates how thoughtful interface design can make government-published data genuinely actionable for everyday users.

5.1 Limitations

- **Static, manually updated price data.** All fuel prices are hardcoded in data.js and sourced from the DOE Philippines weekly retail pump price advisory, published every Tuesday by the Oil Industry Management Bureau (OIMB) via their Oil Monitor portal and Retail Pump Prices & Quality Service Dashboard (doe.gov.ph). However, the application has no mechanism to fetch this automatically — prices go stale the moment a new advisory drops, and updating requires a developer to manually edit the file and redeploy. This is especially critical during the 2026 Philippine energy crisis, where prices have been adjusting multiple times per week. A future version could automate this by consuming DOE data through the Philippine Open Data portal (data.gov.ph), parsing the weekly PDF advisories, or syncing from aggregators like GasWatch PH, which already updates within 24 hours of each DOE release.
- **No persistent storage.** Community price reports submitted by users are not saved anywhere. They trigger a toast notification but are neither stored locally nor sent to any server. Without a backend or cloud database, the community reporting feature is effectively cosmetic and does not contribute to real data accuracy.

- **Third-party API dependency.** The station locator relies on the Overpass API, a third-party OpenStreetMap query service with no guaranteed uptime. During peak usage or server degradation, station queries can fail or time out even with the three-mirror fallback implemented in the code.
- **Limited brand coverage.** Price data covers only 8 major Philippine fuel brands. Independent and local station chains — which are common outside Metro Manila — are not represented, reducing the usefulness of the station comparison feature in provincial areas.
- **No offline support.** The application has no service worker or caching strategy, meaning it requires an active internet connection for maps, location search, routing, and station discovery. A PWA implementation would address this.

6 Conclusion

GasTos PH is a purpose-built web application addressing the practical fuel cost estimation needs of Filipino drivers in a period of record-high pump prices. Through the integration of DOE price data, OpenStreetMap services, and a modular JavaScript architecture, the application delivers calculator, price lookup, station discovery, and community reporting functionality in a single, lightweight, server-free package. Future development directions include automated DOE data ingestion, persistent community reports via a cloud backend, expanded station coverage through additional data sources, and a progressive web app (PWA) implementation for offline use and home screen installation.

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